VELLUM DESKTOP V2.0

RELEASE
v2.0-STABLE

Academic Literature Discovery, Systematic Review Screening & Citation Analysis Workbench

Fernly Botanical Bento Edition

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1. Executive Summary & Core Philosophy

Vellum Desktop V2.0 is a self-contained, privacy-first academic research suite designed for researchers, clinicians, and literature synthesis methodologists. Unlike commercial web platforms that lock bibliographies into proprietary clouds, Vellum operates **100% locally on your machine**. Your research vaults, screening decisions, and PDF libraries never leave your local filesystem.

■ **ASReview Active Learning** Continuous ML literature screening with TF-IDF and Naive Bayes re-ranking. Reduces title/abstract screening effort by up to 70%.	■■ **Connected Papers Graph** Interactive D3 force-directed citation network with seed halo, mint-to-pine temporal gradients, and direct Connected Papers integration.
■ **Visual Boolean Builder** Federated queries across OpenAlex, Crossref, PubMed, and arXiv with structured WHERE / AND / OR clauses and Scopus/RIS ingestion.	■ **Multi-Engine PDF Downloader** Batch paper retrieval with Unpaywall, PubMed Central, and open access resolvers directly into organized project vaults.

2. Quick Start: How to Run in 1 Click

The suite includes a zero-configuration launcher that requires no manual terminal setup or command-line experience.

Step 1: Double-click the launch script located in the extracted folder:

```
CLICK_TO_RUN_VELLUM_DESKTOP.bat   (or 1_launch_desktop_app.bat)
```

Step 2: The script automatically detects the local Python environment and installs any missing packages.

Step 3: Your default browser will launch automatically to the interface:

```
http://localhost:8501
```

Component	Minimum Requirement	Recommended
Operating System	Windows 10 / 11 (64-bit), macOS, Linux	Windows 11 (64-bit)
Python Environment	Python 3.10+ (Included in .venv)	Python 3.11 / 3.13 (.venv pre-configured)
Memory (RAM)	4 GB	8 GB+ (for active learning models)
Web Browser	Microsoft Edge, Google Chrome, Firefox	Chromium-based browser (Edge / Chrome)

3. Module 1: ASReview Active Learning Screening

The **ASReview Active Learning Screening Hub** (available under the ■ Screening tab) brings machine-learning-assisted systematic review screening directly to your desktop. It follows the official protocol developed by researchers at Utrecht University.

Key Screening Capabilities:

- **Multi-Source Ingestion:** Import references from Scopus CSV, Web of Science, PubMed, Covidence RIS, BibTeX, or directly from your active Project Paper Pool.
- **3-Tier Automated Deduplication:** Evaluates incoming papers using Normalized DOI matching, Normalized Title hashing, and Token Jaccard similarity (> 0.88), reporting removed duplicates and preserving metadata.
- **Continuous Priority Re-Ranking:** Every time you Include (I) or Exclude (E) a study, the underlying TF-IDF + Multinomial Naive Bayes model retrains and dynamically surfaces the most promising candidates first.
- **Stopping Heuristic Rules:** Monitors consecutive irrelevant records. When 30–50 consecutive negatives are reached, the system signals that the estimated residual recall plateau exceeds 95%.
- **Session Management:** Create and save independent screening sessions per project (e.g., Primary RCTs, Diagnostic Reviews) with local persistence.
- **1-Click Library Export:** Send all included studies directly to your project's Unified Paper Pool and download queue.

Key Shortcut	Screening Action	Behavior
[I]	■ Include Study	Marks study as relevant, retrains model, re-ranks remaining queue
[E]	■ Exclude Study	Marks study as irrelevant, increments consecutive exclude counter
[U]	■■ Undo Decision	Reverts previous label and restores paper to pending queue
[Skip]	■■ Defer / Skip	Moves paper to back of pending list without updating ML priors

4. Module 2: Connected Papers Scholarly Force Graph

Located in **Discover** → **Citation Network**, this module generates interactive force-directed bibliometric graphs around any seminal seed paper (by DOI or Title).

- **Connected Papers Botanical Styling:** Origin seed paper is highlighted with an unmistakable purple halo (#a855f7). Surrounding literature transitions smoothly from pale mint (foundational prior works) to deep forest pine (contemporary derivative breakthroughs).
- **Citation-Proportional Radii:** Node sizes scale logarithmically based on global citation counts (log10(citations + 1)).
- **Direct Connected Papers Deep-Link:** Click ■■ **Connected Papers** on any selected paper to open its live web graph at https://www.connectedpapers.com/main/{doi}/graph.
- **Seamless Screening Handoff:** Click ■ **Screen in ASReview** on any discovered paper to automatically stage and screen it in your active review session.

5. Literature Search, PDF Downloader & PRISMA 2020

• **Visual Boolean Builder:** Construct multi-clause queries (Title / Abstract, Author, Year) connected with uppercase AND, OR, and NOT operators. Automatic syntax validation prevents common API errors.
• **Unified Paper Pool:** Deduplicated repository of all candidate studies, featuring instant title search, year filtering, and DOI links.
• **Batch PDF Downloader:** Automatically resolves Open Access PDFs via Unpaywall, PubMed Central, and arXiv, saving clean PDF files directly into your project's papers/ vault folder.
• **PRISMA 2020 Flow Export:** Export complete screening statistics (Identified, Deduplicated, Screened, Excluded, Included) to standard PRISMA-compliant CSV and text summaries.

6. Project Directory Structure & Collaboration

When sharing Vellum Desktop V2.0 with colleagues, simply provide the standalone archive. The project structure is organized as follows:

```
PDF_Downloader_Dual_Pack/
  ■■■ dist/
  ■     ■■■ Vellum-Desktop-V2.0-Setup-x64.exe  <-- Windows installer
  ■■■ CLICK_TO_RUN_VELLUM_DESKTOP.bat   <-- 1-Click main launcher
  ■■■ 1_launch_desktop_app.bat          <-- Alternative launcher
  ■■■ README_VELLUM_DESKTOP_V2.0.pdf    <-- This documentation
  ■■■ Desktop_App/
        ■■■ pdf_downloader_app.py       <-- Main application router
        ■■■ vellum_theme.py             <-- Fernly Botanical Bento theme
        ■■■ app_pages/
        ■     ■■■ screening.py          <-- ASReview Active Learning Engine
        ■     ■■■ citation_graph.py     <-- Connected Papers force graph
        ■     ■■■ project_overview.py   <-- Dashboard & PRISMA matrix
        ■     ■■■ library.py            <-- Unified Paper Pool
        ■■■ requirements.txt            <-- Complete dependencies (UTF-8)
        ■■■ build_vellum_windows.spec   <-- PyInstaller build spec
        ■■■ installer.iss               <-- Inno Setup installer script
        ■■■ tests/                      <-- 64 unit tests + page smoke test
```

7. Quality Assurance & Test Verification

Vellum Desktop V2.0 is rigorously tested with automated unit and regression suites:

- **Test Command:** Desktop_App\.venv\Scripts\python.exe -m pytest -o pythonpath=Desktop_App Desktop_App\tests -v
- **Suite Results: 64 passed (100% pass rate)** across deduplication, active learning re-ranking, and graph generation.
- **Page Smoke Test:** Desktop_App\.venv\Scripts\python.exe Desktop_App\tests\smoke_app_pages.py runs the main app and all 10 pages in-process through Streamlit's AppTest framework - **11 / 11 ran clean**, with zero exceptions.
- **Bytecode Integrity:** All modules compile with zero syntax errors via py_compile.
- **Frozen Bundle:** The compiled Vellum.exe answers /_stcore/health with HTTP 200 on 127.0.0.1.